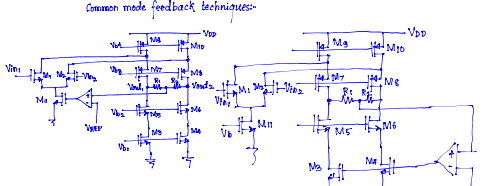
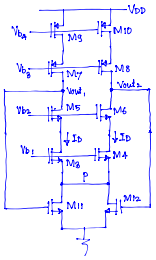
Common mode feedback techniques:-

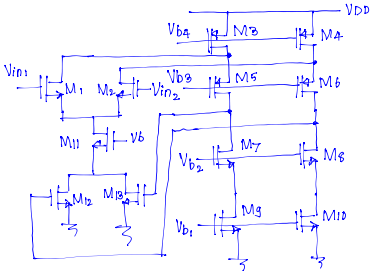


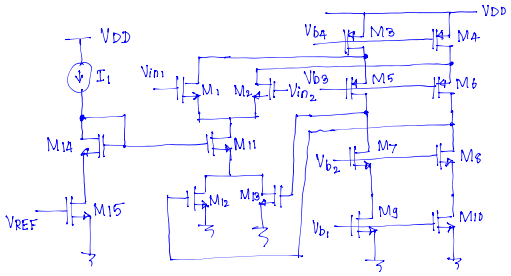


$$V_{b_1} - V_{GS8} = 2I_D R_{total\,p}$$

$$= 2I_D \frac{1}{\mu_n C_{ox}\left(\frac{W}{L}\right)_{11,12}\left(V_{out_1}+V_{out_2}-2V_{th}\right)}$$

$$V_{out_1}+V_{out_2} = \frac{2I_D}{\mu_n C_{ox}\left(\frac{W}{L}\right)_{11,12}} \cdot \frac{1}{V_{b_1}-V_{GS8}} + 2V_{th}$$

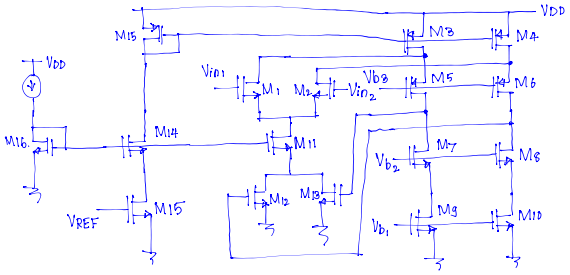




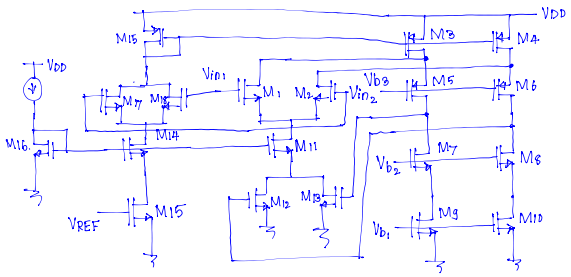
$$I_{D11} = I_{D14} = I_1$$

$$\left(\frac{W}{L}\right)_{14} = \left(\frac{W}{L}\right)_{11}$$

$$\left(\frac{W}{L}\right)_{15} = \left(\frac{W}{L}\right)_{12} + \left(\frac{W}{L}\right)_{13}$$



For removing finite error in I_{D11} and I_{D14} (since $V_{DS11} \neq V_{DS14}$)



$$\left(\frac{W}{L}\right)_{17} = \left(\frac{W}{L}\right)_1$$

$$\left(\frac{W}{L}\right)_{18} = \left(\frac{W}{L}\right)_2$$